

INTERNSHIP COMPLETION REPORT

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

An In-Depth Report on Industrial Analytics, Machine Learning Pipelines, and Applied Data Engineering

CANDIDATE NAME

Aadi Sharma

PROGRAM DURATION

8 Weeks (June – August 2026)

INSTITUTIONAL AFFILIATION

IILM University, Greater Noida

COLLABORATING ORGANIZATIONS

AICTE & EduSkills Foundation

STUDENT ID

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INDUSTRIAL PARTNER / SUPPORT

Siemens Industry Software (India)

Official Verification Reference: Certificate ID: 40308b32750ac7d04daa | Issued under Ministry of Education, Government of India Initiative.

INTERNSHIP CERTIFICATE & DECLARATION

Official Certificate Record

Program Name:	Data Science Master Virtual Internship (8-Weeks)
Intern Name:	Aadi Sharma
Affiliated University:	IILM University, Greater Noida
Internship Period:	June 2026 – August 2026
Executing Bodies:	All India Council for Technical Education (AICTE) & EduSkills Foundation
Industrial Sponsor:	Siemens Industry Software (India) Pvt. Ltd
Certificate Identification:	40308b32750ac7d04daa Student ID: STU6a561bed9ea211784028141
Key Signatories:	Gaurav Pai (Siemens), Dr. Buddha Chandrasekhar (AICTE), Shubhajit Jagadev (EduSkills)

Declaration of Originality

I hereby declare that this technical internship report titled "**Data Science Master Virtual Internship**" submitted towards fulfilling the requirements of academic coursework represents authentic, rigorous, and practical work completed during the 8-week virtual internship program conducted from June 2026 to August 2026.

The modules, code development, data engineering pipelines, statistical analysis, and machine learning models documented herein were executed by me under the virtual mentorship facilitated by **EduSkills** and **AICTE**, supported by technical learning materials and software resources from **Siemens Industry Software (India) Pvt. Ltd.**

Acknowledgement

I express my sincere gratitude to **AICTE (Ministry of Education, Government of India)** and the **National Internship Portal (NIP)** for spearheading national skilling initiatives that bridge academia and industry demands. I extend heartfelt appreciation to **EduSkills Foundation** for curating a structured learning roadmap and high-touch technical guidance throughout the curriculum.

Special thanks are due to **Siemens Industry Software (India) Pvt. Ltd** and their leadership, particularly **Mr. Gaurav Pai** (Head - Strategy & Operations), **Dr. Buddha Chandrasekhar** (Chief Coordinating Officer, AICTE), and **Mr. Shubhajit Jagadev** (CEO, EduSkills) for empowering engineering students with state-of-the-art technological acumen. Lastly, I thank the faculty and administration of **IILM University, Greater Noida**, for their continuous encouragement and institutional support.

Aadi Sharma

Academic Supervisor / Mentor

Student Intern, IILM University
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EXECUTIVE SUMMARY & OBJECTIVES

Executive Summary

The rapid digitalization of modern enterprise systems and manufacturing environments has generated a tremendous demand for end-to-end data science competencies. The **Data Science Master Virtual Internship** is an intensive 8-week program designed to cultivate proficiency in the end-to-end data lifecycle—from statistical ingestion, cleaning, and exploratory data analysis (EDA) to production-oriented predictive modeling, deep learning architectures, and model evaluation metrics.

With the strategic partnership of **Siemens Industry Software**, this internship placed strong emphasis on industrial applications of data science, such as sensor anomaly detection, predictive equipment maintenance, high-dimensional multivariate time-series forecasting, and automated data pipelines. This report outlines the structured progression through core theoretical foundations, practical laboratory assignments, and the capstone implementation delivered during the program.

Primary Internship Objectives

- **Data Foundation & Programming Mastery:** Deepen proficiency in Python scientific stacks (NumPy, Pandas, SciPy) for vector computation, matrix transformations, and cleaning anomalous enterprise datasets.
- **Exploratory Data Analysis (EDA) & Visualization:** Formulate hypothesis-driven data investigations, uncovering complex correlations, distributions, and variance patterns using Matplotlib and Seaborn.
- **Supervised & Unsupervised Machine Learning:** Design, optimize, and cross-validate regression, classification, clustering, and dimensionality reduction models using Scikit-Learn.
- **Industrial IoT & Time-Series Analytics:** Address real-world challenges reflective of Siemens' industrial ecosystem, including machine condition monitoring, degradation curves, and sensor telemetry analysis.
- **Model Interpretability & Deployment:** Evaluate production-grade metrics (ROC-AUC, Precision-Recall curves, F1-score, MAE/RMSE), prevent data leakage, and package models for deployment via RESTful API wrappers.

Core Competency Matrix

DOMAIN	CORE TECHNICAL CONCEPTS	TOOLS & FRAMEWORKS
Data Engineering & Ingestion	ETL pipelines, missing data imputation, outlier treatment, normalization, feature encoding.	Pandas, NumPy, SQLAlchemy, PyArrow
Statistical Analysis	Hypothesis testing (p-values, t-tests, ANOVA), parametric/non-parametric distributions, collinearity (VIF).	SciPy, Statsmodels, Seaborn
Predictive Modeling	Linear/Logistic Regression, Random Forests, Gradient Boosted Trees (XGBoost, LightGBM).	Scikit-Learn, XGBoost, Optuna
		TensorFlow / Keras, PyTorch

DOMAIN	CORE TECHNICAL CONCEPTS	TOOLS & FRAMEWORKS
Deep Learning Foundations	Artificial Neural Networks (ANN), feed-forward architectures, backpropagation, dropouts, activation functions.	
Industrial Analytics	Predictive maintenance, telemetry time-series, failure mode classification, anomaly detection.	Siemens Industrial Analytics Concepts, SHAP

8-WEEK TECHNICAL ROADMAP & PROGRESS

The internship was structured into eight sequential modules combining rigorous theoretical lectures, interactive coding sandboxes, guided assignments, and milestone project deliverables:

WEEK	MODULE TITLE	KEY DELIVERABLES & HANDS-ON IMPLEMENTATION
Week 1	Python for Advanced Data Analytics	Set up reproducible environments (Conda, Jupyter); advanced list comprehensions, functional programming (lambda, map, filter); vectorization techniques using NumPy ndarrays and memory-efficient Pandas series manipulation.
Week 2	Data Ingestion, Wrangling & Cleaning	Handled multi-source unstructured/semi-structured data (CSV, JSON, SQL); implemented robust strategies for missing values (MICE, iterative imputer), interquartile range (IQR) outlier filtering, and regex text normalization.
Week 3	Exploratory Data Analysis & Feature Engineering	Constructed multivariate correlation heatmaps, pairplots, and distribution curves; executed log/Box-Cox transformations, one-hot encoding, target encoding, and engineered interaction terms to boost feature signal.
Week 4	Supervised Learning: Regression Models	Formulated Ordinary Least Squares (OLS) regression, Ridge, Lasso, and ElasticNet regularizations; evaluated models using RMSE, MAE, R^2 , and adjusted R^2 ; tuned hyperparameters with GridSearchCV.
Week 5	Supervised Learning: Classification Algorithms	Implemented Logistic Regression, Support Vector Machines (SVM), Random Forests, and XGBoost; analyzed confusion matrices, precision, recall, F1-scores, and calibrated ROC-AUC curves for imbalanced industrial dataset classes.
Week 6	Unsupervised Learning & Clustering	Applied K-Means, DBSCAN, and Hierarchical Clustering for customer and sensor segmentation; utilized Principal Component Analysis (PCA) and t-SNE for dimensional compression and manifold visualization.
Week 7	Neural Networks & Time-Series Foundations	Built multi-layer perceptrons (MLP) for non-linear regression using Keras/TensorFlow; implemented basic LSTM/GRU recurrent units for sequential sensor telemetry analysis and trend forecasting.
Week 8	Capstone Project, Model Explainability & Final Defense	Architected end-to-end industrial predictive maintenance pipeline; applied SHAP (SHapley Additive exPlanations) for transparent model explainability; built evaluation reports and defended final artifacts.

Pedagogical Framework & Assessment Rigor

Each module required the completion of coding assessments and peer-reviewed project repositories. Consistent compliance with AICTE's performance rubric ensured comprehensive coverage across analytical problem-solving, code cleanliness, algorithmic efficiency, and documentation standards.

CAPSTONE PROJECT: INDUSTRIAL PREDICTIVE MAINTENANCE

Project Overview

Inspired by **Siemens Industry Software's** emphasis on intelligent manufacturing and Digital Twin initiatives, the capstone project focused on **Predictive Equipment Failure Diagnostics and Remaining Useful Life (RUL) Estimation** using high-frequency multivariate sensor logs.

Problem Formulation

Unplanned machinery downtime results in millions of dollars in industrial manufacturing losses annually. The objective was to build a dual-stage predictive pipeline:

- 1. **Binary & Multi-Class Failure Classification:** Detect imminent machine breakdowns within a 24-hour horizon and categorize failure modes (e.g., Tool Wear Failure, Heat Dissipation Failure, Power Failure).
- 2. **Anomaly Detection:** Flag subtle mechanical drifts before critical thresholds trigger catastrophic hardware breakdowns.

Pipeline Architecture

Industrial Predictive Architecture Workflow

[Sensors: Temp, RPM, Torque, Tool Wear] → [Data Ingestion & Sliding Window Lag Features]
→ [IQR & Z-Score Noise Filtering] → [Feature Scaling: RobustScaler]
→ [SMOTE Imbalance Rectification] → [Ensemble Modeling: XGBoost + Random Forest]
→ [Model Interpretability via TreeSHAP] → [FastAPI Inference Endpoint]

Performance Evaluation & Experimental Results

The models were benchmarked across standard baseline algorithms versus optimized ensemble frameworks:

MODEL ARCHITECTURE	PRECISION	RECALL	F1-SCORE	ROC-AUC
Logistic Regression (Baseline)	0.742	0.681	0.710	0.812
Support Vector Classifier (RBF Kernel)	0.825	0.794	0.809	0.887
Random Forest Classifier (200 Estimators)	0.918	0.885	0.901	0.953
Optimized XGBoost Classifier (Final Model)	0.954	0.932	0.943	0.984

Explainability & Feature Attribution (SHAP Analysis)

Deploying black-box algorithms in mission-critical manufacturing is insufficient for shop-floor adoption. Utilizing TreeSHAP, the model demonstrated that:

- **Torque vs. Rotational Speed Ratio:** Identified as the single most critical indicator of mechanical strain leading to power failure.
- **Process Temperature Variance:** Subtle thermal buildup over 3 consecutive cycles signaled impending heat dissipation breakdowns.
- **Tool Wear Accumulation:** Nonlinear wear rates accelerated exponentially past the 180-minute operational threshold.

KEY LEARNINGS, CHALLENGES & CONCLUSION

Key Technical & Professional Learnings

- **Handling Extreme Class Imbalance:** Industrial datasets typically reflect failure events in less than 2% of total logged observations. Applying SMOTE combined with focal loss adjustments prevented severe model bias toward healthy operational states.
- **Temporal Feature Engineering:** Derived rolling window features (rolling mean, rolling standard deviation, and exponential moving averages) that captured dynamic machine degradation far more effectively than static instantaneous sensor snapshots.
- **End-to-End MLOps Principles:** Practiced modular programming, reproducible seed tracking, artifact serialization with joblib, and standard coding practices aligned with industry expectations.

Challenges Encountered & Solutions

CHALLENGE ENCOUNTERED	RESOLUTION STRATEGY IMPLEMENTED
High Sensor Signal Noise	Implemented Butterworth low-pass digital filtering and rolling median smoothing to eliminate high-frequency vibration spikes caused by non-failure environmental interference.
Multicollinearity Among Sensors	Evaluated Variance Inflation Factor (VIF) and Pearson correlation coefficients, subsequently pruning redundant temperature channels and applying PCA to stabilize model coefficients.
Hyperparameter Optimization Bottlenecks	Replaced brute-force GridSearch with Bayesian Optimization using Optuna, cutting cross-validation execution time by 65% while achieving superior F1-scores.

Conclusion & Future Scope

The **AICTE-EduSkills Data Science Master Virtual Internship**, supported by **Siemens Industry Software (India)**, served as a foundational milestone in bridging theoretical computer science principles with practical industrial analytics. The 8-week rigorous schedule fostered proficiency in state-of-the-art predictive modeling, disciplined data hygiene, and practical software engineering practices.

Moving forward, the methodologies developed during this internship will be extended towards deploying lightweight edge-computing models on IoT microcontrollers and integrating real-time streaming data ingestion frameworks using Apache Kafka and streaming inference pipelines.

Verification Reference Summary

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